

Sample Exam 4 Questions/Concepts on Mohr's Circle

Conceptual Questions for Mohr's Circle

1. Can Mohr's Circle be used in determining non-principal moments of inertia?
2. Can Mohr's Circle be used to determine I_{min} and I_{max} ?
3. Theta (θ) degrees of rotation on an area cross section ("cross section" refers to the original shape or area) is represented by what angle of rotation on MC?
4. How do the sense (direction) of the angle on the cross section ("cross section" refers to the original shape or area) compare to the sense (direction) of the angle on MC?
5. For a given area and original set of axes (e.g. x-y), can multiple sets of points be plotted on MC?
6. What values are plotted on the MC horizontal axis?
7. What values are plotted on the MC vertical axis?
8. For a given non-symmetric area and origin, how many sets of principal axes are there?
9. For a symmetric area and origin (original set of axes), what is the angle to the nearest principal axis?
10. I_{max} and I_{min} are "principal" axes, not "principle" axes. (Know how to spell these.)
11. Principal moment reference points are located where the circumference of Mohr's Circle intersects the I axis.
12. The center of Mohr's Circle is always halfway between the principal moment of inertia reference points.
13. Regarding rotated axes, the major principal axis has the most area concentrated farthest from it.
14. Any axis rotated from the major principal axis will have a smaller moment of inertia.
15. Product of Inertia about the principal axes will always be zero.
16. On MC, two points that lie on either end of a diameter line represent the MOI and POI of a set of axes that are perpendicular to each other on the area cross-section.
17. Points $[I_x, I_{xy}]$ or $[I_u, I_{uv}]$ plotted on the circle are sometimes called "reference points," especially when used to construct the circle.
18. Is I_{MIN} or I_{MAX} ever negative?

Conceptual Questions Involving Pictures for MC

1. All of the axes shown pass through the origin and only one of them is principal. Identify the minor principal axis.
A. 1
B. 2
C. 3
D. 4
E. 5
F. 6
2. All of the axes shown pass through the origin and only one of them is principal. Identify the minor principal axis.
A. 1
B. 2
C. 3
D. 4
E. 5
F. 6

